

Official Literature Search

I. Significance Statement

AMR is increasingly recognized as a major public health concern in Canada and globally, with an estimated 4.71 million deaths associated with bacterial AMR worldwide in 2021 (Naghavi et al.). While AMR is a global problem, it can be shaped by local healthcare structures and policy-making. In Canada, AMR responses reportedly have been fragmented and decentralized provincially, limiting a coordinated national action plan. During the COVID-19 pandemic, resources and attention were greatly redirected, which may have caused AMR responses to not be prioritized (Lorcy et al., 2026). Additionally, increased cases of bacterial pneumonia caused by COVID-19 were combatted with poor antibiotic stewardship, leading to a surge of antimicrobial resistance.

II. What is Antimicrobial Resistance?

Antimicrobial resistance (AMR) is a pervasive medical phenomenon that greatly reduces the efficacy of antibiotics in treating bacterial infections, affecting millions of people worldwide. Resistance can arise through a variety of mechanisms, including enzymatic antibiotic degradation, reduced membrane permeability, and efflux pumps that remove antibiotics from bacterial cells (Abbas et al.) Instead of bacteria spontaneously “developing” resistance, AMR occurs when naturally resistant bacterial variants, or those who acquired resistance through genetic exchange mechanisms, survive antibiotic exposure while susceptible bacteria are inhibited. This creates a selective pressure that allows resistant variants to spread. These variants may occur naturally, but their prevalence and clinical impact have been amplified by human antibiotic use for disease treatment.

AMR is a highly multifaceted global health issue that can be influenced not just by antibiotic misuse but also by biological, environmental, and socioeconomic factors within a “sociobiome” (Hinchliffe et al., 2018). It is important to understand that resistance mechanisms can arise naturally through evolution and genetic mutations, however, such factors work by exacerbating and accelerating their development and spread.

IV. Risk Factors

Risk factors contributing to AMR are given rise to by a variety of factors such as healthcare practices, environmental conditions, and socioeconomic status. Interestingly, the COVID-19 pandemic appears to have exacerbated AMR globally, with CDC reporting a significant increase in carbapenem-resistant *Acinetobacter* and *Enterobacterales* in U.S. hospitals. While the underlying causes are not yet fully understood, possible drivers could include inappropriate antibiotic use at the start of the pandemic and prolonged ventilator and catheter use for the care of critically ill COVID-19 patients. Also during the pandemic, a major epidemiological shift was observed in Vancouver in multidrug-resistant *Shigella sonnei* infections before and after the pandemic, where infections became increasingly associated with people experiencing homelessness rather than through sexual transmission, which was the primary route previously. Suggested contributing factors include malnutrition, poor sanitation access, overcrowded living

conditions, and substance-use disorder may lead to worsening overall health and increased vulnerability to infection.

III. Who is affected?

Among these socioeconomic and social aspects, it becomes important to consider what clusters of patients are differentially affected by AMR. Antibiotic use patterns reveal significant disparities across demographic and socioeconomic groups, with the highest utilization rates observed among individuals with pre-existing medical conditions (46.1%), young adults aged 18-24 (46.2%) and 25-34 (36.3%), those with household incomes below \$60,000 (38.2%), individuals without university education (37.9-38.8%), and women compared to men (37.3% vs. 29.7%) (Abbas et al., 2024). These patterns of elevated antibiotic exposure in already vulnerable populations compound their risk for both developing and encountering antimicrobial resistance, creating a troubling intersection between health inequity and AMR burden that warrants closer examination of specific high-risk groups.

Cancer Patients

One such vulnerable population is cancer patients, who are disproportionately affected by antimicrobial resistance due to frequent hospitalizations, prior antibiotic use creating selection pressure, immunocompromised status, and medical interventions like surgeries and urinary catheters. A systematic review found that 47% of studies examining AMR in cancer patients reported mortality as an outcome, highlighting the severe clinical impact on this immunocompromised population (Ntim et al., 2025). Among the bacterial pathogens studied, *E. coli* demonstrated particularly concerning resistance patterns, with 82% of isolates resistant to penicillins and approximately 50% of isolates producing extended-spectrum beta-lactamases (ESBLs), a major multi-drug resistance mechanism that breaks down beta-lactam antibiotics. Furthermore, resistance patterns varied significantly by cancer type, with solid tumors showing higher resistance to penicillins and tetracyclines, while blood-related issues demonstrated elevated resistance to cephalosporins and vancomycin.

Long-term Care Facilities

Beyond cancer patients, individuals with chronic comorbidities in long-term acute-care rehabilitation hospitals represent another critically vulnerable population facing disproportionate AMR risk. These facilities, which care for patients with conditions such as autoimmune disorders, psychiatric disorders, and severe brain injuries, create ideal conditions for AMR proliferation due to prolonged patient stays (often exceeding one year), extensive medical device use, frequent antibiotic exposure, and challenges in implementing antimicrobial stewardship programs with limited resources. Notably, patients with severe acute brain injury face the greatest risk of healthcare-associated infections from AMR bacteria due to their prolonged rehabilitation needs, multiple comorbidities, invasive procedures, and reliance on medical devices (Grifoni et al., 2025). High prevalence rates of MRSA, VRE, and ESBL-producing bacteria have been consistently documented in these settings, underscoring the heightened AMR burden in long-term care environments.

ICU Nurses

The challenge of AMR transmission extends beyond patients to the healthcare workers who care for them. A recent multicentre cross-sectional study of 290 ICU workers across eight Chinese medical centers revealed that ICU healthcare workers exhibited significantly higher total antibiotic resistance gene (ARG) abundance and diversity compared to healthy controls, with 243 ARGs upregulated in their gut microbiota (Huang et al., 2025). Nursing assistants demonstrated the highest total ARG abundance, with resistance genes primarily originating from *Enterococcus faecium*, *Klebsiella pneumoniae*, and *Enterobacter* species. This elevated ARG carriage likely results from unavoidable occupational exposure to ICU environments where air concentrations of ARGs exceed external environments and healthcare surfaces are heavily colonized by multidrug-resistant organisms, creating potential bidirectional transmission pathways between healthcare workers and vulnerable patient populations.

Indigenous Populations

Beyond healthcare settings and worker populations, Indigenous communities face a particularly severe and systemically driven AMR burden rooted in colonial healthcare structures and ongoing inequities. A retrospective and prospective observational study conducted at Alice Springs Hospital in Central Australia revealed that Indigenous patients experienced superficial skin and soft tissue infections (SSTIs) at a rate 6.6 times higher than non-Indigenous patients (18.9 vs. 2.9 per 1,000 person-years). Community-associated methicillin-resistant *Staphylococcus aureus* (CA-MRSA) predominated at 57% of isolates and 90% carrying the Panton-Valentine leucocidin (PVL) toxin gene, a virulence factor strongly associated with severe, recurrent infections. More importantly, 36.7% of patients received empiric antimicrobials that were inactive against their pathogen, with flucloxacillin (ineffective against MRSA) prescribed in 50% of cases, highlighting a substantial gap between prescribing practices and actual resistance profiles (Harch et al., 2017). These disproportionate outcomes can be explained by structural determinants, including underfunded and overcrowded housing, inadequate access to sanitation and clean water, geographic isolation in remote communities, shortages of healthcare professionals, limited diagnostic testing capacity, and substantial barriers to medication access. Such conditions increase infection risk and reliance on empirical antibiotic treatment while simultaneously impeding proper infection management and antimicrobial stewardship.

Current AMR governance frameworks in Canada often operate under one-size-fits-all assumptions of stable healthcare access, specialist availability, consistent diagnostics, and urban infrastructure that do not align with Indigenous lived experiences, resulting in policies that fail to reflect the realities of remote and underserved communities. Effective AMR strategies must move beyond deficit-based approaches that portray Indigenous communities as vulnerable or poorly educated. One paper suggests a better approach would be building upon existing community strengths such as collective caregiving, local leadership, and demonstrated resilience and adaptability through Indigenous-led, culturally safe communication that centers trust-building, incorporates Indigenous knowledge systems, and ensures meaningful representation in policymaking (Hindmarch et al., 2024).

Homeless Individuals

People experiencing homelessness represent a critically underserved population whose structural vulnerabilities create compounding AMR risks that are difficult to address through conventional stewardship frameworks. A striking illustration of this emerged in Vancouver, where a dramatic epidemiological shift in multidrug-resistant *Shigella sonnei* infections was documented

following the COVID-19 pandemic. Between 2015 and 2020, MDR *S. sonnei* infections were predominantly associated with men who have sex with men, with only 6% of cases occurring among homeless individuals. By 2021 to 2022, this pattern had reversed markedly: 77% of patients were experiencing homelessness and 82% presented with comorbid substance use disorder, with infections being more severe and rates of hospitalization substantially higher (Stefanovic et al., 2025). The outbreak strain demonstrated resistance to ciprofloxacin, azithromycin, ampicillin, and trimethoprim, severely narrowing the available treatment options for this population.

Several intersecting structural factors underpin this shift. Substance use disorder emerged as a particularly significant driver, worsening overall health, increasing vulnerability to infection, and elevating hospitalization risk. Malnutrition and limited sanitation access further compound biological susceptibility, while unstable living conditions create persistent barriers to timely healthcare-seeking and consistent treatment adherence. Critically, the shift from sexual to environmental transmission reflects the physical realities of congregate shelter living, where poor hygiene infrastructure and overcrowding facilitate widespread interpersonal spread of resistant organisms (Stefanovic et al., 2025). Proposed interventions address these structural roots directly, including expanded access to sanitation and handwashing facilities, improved nutritional support, and vaccination programs targeting *Shigella*.

Refugees and Asylum Seekers

Forcibly displaced populations carry a disproportionate and frequently underestimated AMR burden, shaped by the compounding vulnerabilities of displacement itself. In 2015 alone, 65.3 million people were forcibly displaced worldwide, a quarter of whom were classified as refugees, yet the AMR research literature has been slow to account for their unique exposure pathways (de Smalen et al., 2017). A systematic review of 577 articles, from which 17 met inclusion criteria, found that *Mycobacterium tuberculosis* was the dominant resistant organism, appearing in 65% of included studies, with multidrug-resistant TB reported in 70% of those cases. Notably, 70% of TB cases were identified through active screening programs rather than clinical hospital admission, suggesting that the true burden would go largely undetected without systematic surveillance. Refugees and asylum seekers were found to carry a relative risk of 4.0 for contracting or reactivating MDR-TB compared to host country populations, a figure that reflects not inherent biological susceptibility but the structural conditions of displacement (de Smalen et al., 2017).

Beyond tuberculosis, multidrug-resistant Enterobacteriaceae cases were documented among paediatric refugee patients in Germany, with resistance spanning penicillins, cephalosporins, and fluoroquinolones. In Austria, *Shigella* cases demonstrated resistance to azithromycin, the recommended first-line treatment for *Shigella* in children, while cases in Greece exhibited simultaneous resistance to between three and eight antibiotic classes. One particularly severe case involved a Libyan refugee transferred to Italy whose MDR *Klebsiella pneumoniae*, resistant even to carbapenems as a last-resort antibiotic class, had been acquired across multiple healthcare systems during treatment of war injuries, illustrating how fragmented cross-border care can itself become a driver of resistance accumulation (de Smalen et al., 2017). The crowded conditions of refugee camps, lack of consistent medical care, and reliance on self-medication with unregulated antibiotics were identified as the primary structural drivers of AMR spread in this population. The authors recommend compulsory multidrug-resistant organism screening of newly arrived refugees as a foundational preventive measure, while also acknowledging a

significant geographical limitation in the evidence base: despite the majority of the world's refugees residing in low- and middle-income countries, 59% of AMR occurrence in the reviewed literature was reported in Europe, pointing to a serious gap in research representation.

Neonates

Preterm neonates represent one of the most biologically vulnerable populations in the AMR landscape, facing resistance risks that begin before birth and are shaped by a cascade of prenatal, intrapartum, and postnatal exposures. Intestinal and immune immaturity, combined with reduced microbial diversity in the gut, leaves preterm infants highly susceptible to sepsis originating either intrapartum or antenatally (Ojeda et al., 2024). Critically, antibiotic resistance genes have been detected in neonates who never received direct antibiotic treatment themselves, acquired instead through maternal antibiotic use during pregnancy and via prolonged rupture of membranes, demonstrating that AMR exposure in this population begins well before any postnatal clinical decision is made.

The REASON study, which randomized neonates to receive or withhold antibiotics after the first 48 hours of birth and used metagenomic sequencing to quantify antibiotic resistance genes in stool samples, found that unnecessary antibiotic administration did not improve neonatal survival outcomes, yet meaningfully increased AMR risk. Administration of dual-combination antibiotics was associated with an increase in bacterial families harbouring many of the pathogens that drive AMR, while prolonged antibiotic exposure was associated with reduced overall microbial diversity, itself a recognized precondition for heightened resistance risk (Ojeda et al., 2024). Among neonates who received antibiotics, 72% showed greater abundance of genes conferring resistance to beta-lactam antibiotics, and tetracycline resistance increased over time in treated infants. Furthermore, 33% of infants born from mothers with prolonged membrane rupture lasting more than 18 hours showed higher antibiotic resistance gene diversity than those born after shorter ruptures, underscoring maternal health management as a critical point of intervention. These findings call for enhanced antenatal care strategies, including earlier microbial screening and approaches to reducing prolonged rupture of membranes, to limit downstream transmission of resistance genes to preterm infants (Ojeda et al., 2024).

Elderly

Elderly populations, particularly those residing in institutional care settings, face a markedly elevated AMR burden compared to their community-dwelling counterparts. A retrospective observational study analyzing over 20,000 positive urine samples collected from patients aged 65 and older in northeastern France between 2014 and 2017 found that nursing home residents faced a 40% higher risk of antibiotic-resistant Enterobacteriaceae than community-dwelling individuals of the same age group (Pulcini et al., 2019). Resistance was assessed across multiple organisms, including *Escherichia coli*, *Proteus mirabilis*, and *Klebsiella pneumoniae*, tested against a broad panel of antibiotics, including fluoroquinolones, trimethoprim-sulfamethoxazole, and ceftriaxone, as well as for ESBL production. The disproportionate resistance burden in nursing home residents reflects the structural conditions of institutional care: residents are frequently exposed to repeated antibiotic courses, are more likely to have urinary catheters, which were found predominantly among nursing home residents in this study, and are managed in environments where consistent antimicrobial stewardship is difficult to implement due to limited specialist availability and resource constraints (Pulcini et al., 2019).

Across these populations examined, people experiencing homelessness, refugees and asylum seekers, neonates, and elderly individuals in institutional care, a consistent and troubling pattern emerges. In each case, AMR risk is not simply a product of individual behaviour but is fundamentally shaped by structural conditions: displacement, poverty, inadequate sanitation, institutional exposure, and fragmented healthcare access. Rural and remote communities, urban slums, ethnic and racial minorities, and immigrant or refugee populations each experience unique manifestations of systemic exclusion, where factors such as socioeconomic status, geographic location, immigration status, and disability compound AMR vulnerability in ways that existing surveillance systems and policy frameworks remain ill-equipped to fully capture or address (Adebisi & Ogunkola, 2023). It is now clearer than ever that society faces a cycle in which those most vulnerable to infection are simultaneously most exposed to antimicrobial resistance.

V. Persisting Challenges for AMR Patients

In addition to understanding the vulnerable populations disproportionately impacted by AMR, one must also understand that the challenges that patients with AMR face goes beyond medicine alone. To fully understand how patients struggle with AMR, the socioeconomic, social, psychological and systematic dimensions of it need to be unpacked.

Cost and Access Barriers

For instance, studies have postulated that access barriers and cost pressures exacerbate antibiotic abuse (Crago et al., 2022; Abbas et al., 2024). When accessing medical care is time consuming or too expensive, people push for prescriptions as they cannot easily return for follow up care, especially in remote or Northern communities where accessing healthcare is not convenient. Cost pressure is another issue driving unhealthy antibiotic stewardship. High prescription costs push people to share antibiotics and purchase antibiotics abroad over-the-counter for future use. These issues are especially prevalent in Indigenous communities, where they keep leftover antibiotics due to costs and difficulties accessing doctors.

Shared Rehab Spaces

Another limitation is the lack of control over shared rehab spaces. Shared spaces, such as gyms and pools make patient isolation difficult, increasing transmission risk. Furthermore, the lack of on-site labs delays diagnosis of conditions. This gap could be addressed by using rapid diagnostic molecular tests, reducing unnecessary antibiotic prescribing to combat AMR. Meanwhile, caregivers and family members who interact with patients are not trained in infection prevention and control practices, spurring further chances of transmission. In addition, antimicrobial stewardship is weak in these settings due to lack of specialists and limited resources, leading to inappropriate antibiotic prescribing and usage that the authors argue can be tackled by incorporating antimicrobial stewardship programs to promote informed and responsible prescribing.

Knowledge Gaps

Another persisting problem is a lack of knowledge on a variety of levels: healthcare and public levels. Within the literature, it has been underlined that public knowledge gaps is one of the ways through which antibiotic misuse is reinforced. For example, while 81% of Canadians correctly

identified antibiotics as bacteria-killing, some believed antibiotics work against viruses, colds and flu, or fungal infections (Crago et al, 2022). Misinformation was concentrated among young adults of ages 18-24, those with a high school degree or less (~50%), and non-English speaking households. Furthermore, awareness of AMR terminology was lacking, with most familiar with the terms antibiotic resistance, drug resistance, and superbugs.

This information is in-line with what we learned from an integrated human practices (iHP) meeting we had with **Dr. Hancock**, Canada Research Chair in Health and Genomics and a professor of microbiology and immunology. One of the first things Dr. Hancock mentioned to us was that we were misusing the term “antibiotic resistance,” whereas the more accepted term in the literature is “antimicrobial use.” The fact that this term did not show up in the aforementioned study at the very least supports the lack of education among a considerable population of individuals. Moreover, one of the questions we asked Dr. Hancock was about misconceptions regarding AMR, and these are the main points he mentioned:

1. AMR is not a “thing.” It is a collection of things and there are a variety of mechanisms for it.
2. Many people do not understand the concept of resistance; 97% of time antibiotics do work, but they work because physicians have backup plans.
3. Antibiotics are the single most important and successful medicine of all time. Nothing has given the impact that antibiotics has (e.g. there would be no transplantations or surgeries, simple burns and wounds could become septic, pre-term babies would die of infection)

In support of this, patients living with AMR must navigate through many different obstacles. Clinicians themselves lack sufficient knowledge to instruct patients, exacerbating anxiety (Hamilton et al., 2024). Furthermore, even if patients wish to learn more on their own, they cannot easily access information. The lack of support systems cause patients to feel isolated and unsupported, along with low public awareness leading to social stigmas, exacerbating this situation. Proper treatments tend to be more expensive and toxic, worsening the burden put on patients. Moreover, patients can be asymptomatic carriers, increasing the risk of transmission and future infections.

To make this even more evident, an independent line of evidence points to the lack of patient understanding of AMR once again, where patients reported not understanding what AMR was, as clinicians did not explain it to them properly (Whittaker et al., 2023). They were also not informed on what causes antimicrobial resistance and some never heard the term “antimicrobial resistance” even after years of treatment, highlighting the need for better patient communication. Furthermore, multiple patients struggled to find professional advice and access effective treatments, navigating the healthcare system unsupported. Poor communication, lack of educational outreach, and recurring infections worsen uncertainty in patients and public stigma.

All of this comes down to support what Hinchliffe et al. has raised: Current AMR strategies that rely on education and regulation are *insufficient* because it assumes changing behaviour is easy. In reality, antibiotic usage is shaped by economic and environmental factors as well, thus a one-size-fits-all solution fails to address these various factors.

VI. The Psychosocial Barriers of AMR

In addition to educational discrepancies with regard to knowledge of AMR in patients, their families and sometimes physicians, there is also a more subtle, yet draining struggle undertaken by AMR patients: Psychosocial stressors. AMR affects how individuals perceive themselves, interact with healthcare systems and society, and seek support during illness.

A key theme from a qualitative study that interviewed 9 patients living with AMR is that AMR is not only experienced as a medical condition, but also as an ongoing psychological and social burden for the patient and their caregivers (Hamilton et al., 2024). Patients found emotional strains from uncertainty and long-term implications of their illness.

Alongside AMR's burdens from its chronic nature, fear and stigma also become persistent in daily life as patients feel judged by society (Whittaker et al., 2023). This, paired with frustration with local healthcare systems can lead to patients seeking alternative treatments abroad. In addition to this, while other chronic illness patients like cancer or HIV survivors usually have a strong identity and community formed, AMR patients were reported to not identify with a shared "AMR survivor" identity, possibly due to a variety of differences in pathogens and treatment plans (biosociality limitations). This may lead to weak collective advocacy and sense of community, which only further worsens an individual's burdens.

To add to these points, a different study found similar results, where antibiotic stewardship is shaped by a variety of psychological, cognitive and interpersonal barriers that influence both patient and clinician (Donisi et al., 2019). Key psychosocial barriers were identified that could influence an individual's ability to follow antimicrobial stewardship principles, such as low self-efficacy, limited agreement with policies, modest expectations regarding treatment outcomes, and poor motivation to change established habits. Together, these factors may work to limit how well patients follow their treatment plans, especially when facing prolonged, chronic illness.

In addition to these drivers, cognitive and interpersonal factors in clinical settings can further shape how patients navigate their treatment journeys through their healthcare providers. From the clinician side, decision-making in prescribing can be influenced by fear of adverse side effects, uncertainty regarding treatment effectiveness, and reputational risk when going against established practices (Donisi et al., 2019). These influence adherence to clinical practice guidelines.

Lastly, the psychosocial impacts of AMR extend beyond patients themselves and can also affect caregivers and healthcare workers who are involved in their treatment. A qualitative study involving healthcare professionals in six hospitals in Germany found that multidrug-resistant organisms (MDROs) and infection-control procedures were associated with having significant emotional and social consequences (Bushuven et al., 2021). Participants reported that isolation precautions in hospitals, while necessary for reducing bacterial transmission, often lead to feelings of loneliness, abandonment, stigma, and emotional suffering in patients. Some patients described feeling "dirty", contaminated, and ashamed because of their infection status, while healthcare workers observed that family members and visitors were hesitant to interact with isolated patients because of fears regarding transmission. Inconsistent communication and a lack of information regarding MDROs were also identified as contributing to anxiety and fear levels among patients and caregivers.

The study further found that these challenges extend to healthcare workers themselves, who frequently reported feelings of frustration, helplessness, guilt, and compassion fatigue when

caring for patients with MDROs. Participants also described ethical tensions between protecting public health and themselves through infection-control measures and PPE while also preserving patient autonomy and well-being. Together, these findings suggest that the psychosocial burden of AMR is not limited to the infection itself, but also by the social consequences of the healthcare systems where patients receive care.

To mitigate some of these barriers, this research suggests implementing antimicrobial stewardship (AMS) interventions built between AMS experts and prescribers (Donisi et al., 2019). The proposed framework emphasizes first assessing the epidemiological and clinical characteristics of the specific healthcare setting, as well as identifying local psychosocial barriers that may influence antimicrobial prescribing behaviour. Methods like interviews and focus groups were suggested as ways to better understand these factors.

Based on these findings, interventions can be designed using models like the “behaviour change wheel” approach to guide structured strategies and actions targeting factors influencing inappropriate antimicrobial behaviour and the psychosocial determinants. In addition to this, the study also emphasizes the importance of communication and education strategies between AMS teams and clinicians, including training in clinical decision-making, alongside quality control follow-ups and auditing processes to monitor patient outcomes to improve long-term sustainability of stewardship interventions.

Antimicrobial stewardship needs to be balanced two competing needs: ensuring the best chance of effective treatment while mitigating the risk that treatment selects for resistant bacteria at both hospital and community levels. However, achieving this balance is complicated by psychological and cognitive barriers among prescribers. To address these barriers, a structured intervention built around an AMS team across four phases is proposed. On a final note, it is important to understand that AMR is an equity problem rooted deeply within systems. AMR affects the most vulnerable, yet the interventions designed to fight AMR crush the people with least resources. Furthermore, the burden of healthcare associated infections alludes that AMR isn't simply about individual antibiotics misuse, but rather what happens inside healthcare systems themselves. Most AMR data tends to come from hospital isolates and is selection-biased, systematically underrepresenting populations that are most affected, such as caregivers of patients.

Antimicrobial Resistance in Canada

With AMR on the rise in a number of countries, public awareness and perceptions about antimicrobial resistance have become important in shaping stewardship efforts. A Canada-based survey found that 57.5% of respondents expressed concern regarding AMR, with 16% reporting they were “very worried” about the issue (Crago et al.). However, levels of concern were notably lower than those in the United States and the United Kingdom, suggesting possible differences in public awareness. Furthermore, 22% of respondents reported that they or someone they knew had experienced “antibiotic resistance.”

There have been efforts in Canada to tackle AMR use, and this is highlighted by Saatchi et al. who performed a retrospective cohort study on over 26 million BC residents across 18 years. This study found that while overall antibiotic prescribing decreased by 23% between 2000 and 2018, significant disparities persisted across age groups, with adults aged 80 and older receiving antibiotics at rates more than double those of younger adults and showing no decline over the

study period (Saatchi et al., 2021). This elevated prescribing among elderly populations is particularly concerning given their increased vulnerability to adverse antibiotic events, Clostridioides difficile infections, and AMR development, compounded by frequent comorbidities and reduced physiological resilience. Furthermore, upper respiratory tract infections remained the most commonly prescribed indication despite their predominantly viral etiology, indicating persistent inappropriate prescribing patterns that contribute to selection pressure for resistant organisms in the BC community. Although the province's Do Bugs Need Drugs stewardship program has produced measurable reductions in pediatric prescribing (up to 60% decline in children aged 0-2), the failure to curb prescribing among the elderly creates a localized barrier where BC's most vulnerable patients face the dual burden of being both the highest consumers of antibiotics and most susceptible to resistant infections. A secondary finding from the study showed AMR costing the Canadian healthcare system \$1.4 billion in 2018 alone.

Saatchi, A., Morris, A. M., Patrick, D. M., McCormack, J., Reyes, R. C., Morehouse, P., Reid, J., Shariff, S., Povitz, M., Silverman, M., & Marra, F. (2021). Outpatient antibiotic use in British Columbia, Canada: reviewing major trends since 2000. *JAC-Antimicrobial Resistance*, 3(3). <https://doi.org/10.1093/jacamr/dlab116>

Study Design

Retrospective cohort study using population-level administrative health data from British Columbia, Canada, covering a **19-year period (2000-2018)**. The study analyzed data from BC PharmaNet (a centralized prescription database) linked to the Medical Service Plan (MSP) billing system to capture both prescriptions and associated diagnoses.

Methods:

- Antibiotics limited to **oral, acute use** (<30 day supply)
- Prescriptions linked to diagnoses using a validated three-tiered hierarchy system (tier 1: always needs antibiotics; tier 2: sometimes; tier 3: unnecessary)
- Analyzed across three stewardship eras to evaluate the **Do Bugs Need Drugs (DBND)** Community Antimicrobial Stewardship (CAS) program

Three Stewardship Eras:

- Pre-stewardship (2000-2004): Before formal intervention
- Era I (2005-2013): Initial program focused on URTIs
- Era II (2014-2018): Expanded focus on UTI prescribing

Key Populations

- 20+ million unique BC residents included
- 51,367,938 total oral antibiotic prescriptions dispensed over 19 years
- BC population grew from 4.03 million (2000) to 5.00 million (2018)
- Average ~31.1% of BC residents prescribed antibiotics annually
- Most commonly prescribed to **older females in urban settings**
- No differences observed across income quintiles
- Stratified by 7 age groups (0-2, 3-9, 10-18, 19-49, 50-64, 65-79, ≥80 years)

Main Findings

Overall Trends:

- **23% decrease** in outpatient antibiotic prescribing over the study period (652 to 503 prescriptions per 1,000 population)
- Decline observed across all major antibiotic classes **except** "other antibacterials" (driven by nitrofurantoin, which increased 97%)

By Antibiotic Class:

- Penicillins: ↓27% (amoxicillin decreased 25%; cloxacillin ↓90.5%)
- Macrolides: ↓43% overall (but azithromycin ↑146.2%, replacing erythromycin which ↓97%)
- Sulphonamides/trimethoprim: ↓58%
- Quinolones^{**:**} ↓22%
- **Other β-lactams**: ↓12%
- Tetracyclines: Stable overall (doxycycline ↑94.4% offsetting declines)
- Other antibacterials: ↑97% (driven by nitrofurantoin for UTIs)

By Age (Striking Disparities):

- **Children 0-2 years**: ↓60% (largest decline)
- Children 3-9 years: ↓51%
- Children 10-18 years: ↓47.2%
- Adults 19-49 years^{**:**} ↓30.5%
- **Adults 50-64 years**: ↑8.1% (increase)
- Adults 65-79 years: ↓10.6%
- **Adults ≥80 years**: Remained steady at highly elevated rates (1,021-1,178 prescriptions per 1,000 population)

By Indication:

- Upper Respiratory Tract Infections (URTIs): Most prescribed category despite viral etiology; ↓49% but remained the highest indication
- **Urinary Tract Infections (UTIs)**: Second most common; ↑10%
- Lower Respiratory Tract Infections (LRTIs): ↑50% (but low magnitude)
- Acute Otitis Media (AOM): ↓71% (largest decline)
- Skin/Soft Tissue Infections (SSTIs): ↓24%

Stewardship Program Impact:

- **Era I**: 4% overall decline (RR 0.96, 95% CI 0.96-0.97)
- **Era II**: 24% overall decline (RR 0.76, 95% CI 0.76-0.76)
- Most pronounced effect in children aged 0-2 (54% reduction by Era II)
- **Concerning finding**: Adults ≥80 years showed INCREASED prescribing in both eras (Era I: RR 1.17; Era II: RR 1.12)

Key Takeaways:

1. **Stewardship works:** BC's community antimicrobial stewardship program (DBND) is associated with significant reductions in antibiotic use, particularly in children
2. **Age-based disparities persist:** Pediatric populations showed dramatic improvements while elderly populations (especially ≥ 80 years) continue to receive antibiotics at very high rates
3. **URTIs remain problematic:** Despite being predominantly viral and self-limiting, URTIs remain the most prescribed indication, indicating ongoing inappropriate prescribing
4. **Class substitution observed:** While overall use decreased, broader-spectrum agents (e.g., azithromycin) replaced narrower-spectrum first-generation drugs in some classes
5. **Economic burden:** AMR cost the Canadian healthcare system \$1.4 billion in 2018, underscoring the importance of continued stewardship

Implications:

- Provincial stewardship interventions can produce measurable population-level changes
- **Urgent need to target elderly populations** (≥ 65 years) for stewardship efforts, particularly those ≥ 80 years
- Quality of prescriptions (appropriateness) requires further investigation beyond quantity
- Findings position BC as having some of the most conservative antibiotic use rates in Canada

However, despite these initiatives, there are major gaps in understanding what psychosocial barriers patients within British Columbia face given that there is limited research on patient-oriented studies within the province. Moreover, on a more global scale, it is clear that there is a serious need for comprehensive educational initiatives to teach the general public about what AMR is, what it entails and what patients can expect from such a diagnosis.

What we will do at iGEM

Both our iHP meetings and literature search informed what we ended up doing for our inclusivity projects.

Education

To begin, our meeting with Dr. Hancock made it very clear that we needed to have a systematic approach to educating the younger population as much as we could. The postulations of Hinchliffe et al. coupled with the misconceptions raised by Dr. Hancock point to the fact that current efforts towards educating the public are insufficient, so we wanted to do our part by helping educate the younger generation. Namely, we wanted to reframe public understanding and emphasize that:

1. Antimicrobial resistance is the appropriate term to use
2. Antibiotics work because physicians have appropriate plans and dosages for how much antibiotics to prescribe. If not taken to the full, for example, residue bacteria that are resistant to the antibiotics would remain and more complicated treatment plans would be needed to mitigate them.

3. Different populations are differentially at risk of AMR. This was emphasized by our literature search and the different groups that are affected, and Dr. Hancock also noted that individuals who use catheters or are on wheelchairs can become infected more easily and ultimately are more prone to becoming bed-ridden due to their existing low mobility levels.
4. People do not understand how important antibiotics are. As said by Dr. Hancock, antibiotics are the single most important and successful medicine of all time. Nothing has given the impact that antibiotics has (e.g. there would be no transplantations or surgeries, simple burns and wounds could become septic, pre-term babies would die of infection)

To achieve this, we partnered with UBC Geering Up....

The Psychosocial Barriers of AMR Population Clusters- iHP

Lastly, from the clusters of population we came across in our literature search, it is evident that healthcare workers, marginalized patients and those with comorbidities are common groups most disproportionately affected by AMR. Hence, an interesting angle to further dissect is the psychosocial barriers patients with AMR face amongst these populations, and our goal was to do this by interviewing a variety of physicians and nurses regarding the mental health challenges their patients face...

To tie this into our educational efforts, we interviewed patients and clinicians and shared their insight on our social media platforms to make this information available to the general public.